

LAB - 6

Part A Programs

Problem 1 — Display a Character

```
DATA SEGMENT
DATA ENDS
CODE SEGMENT
ASSUME CS:CODE, DS:DATA
START:
    MOV AX,DATA
    MOV DS,AX
    MOV AH,02H
    MOV DL, 'A'
    INT 21h
    MOV AH,4CH
    INT 21H
CODE ENDS
END START
```

```
-t
AX=0744 BX=0000 CX=000F DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0734 ES=0734 SS=0743 CS=0744 IP=0003 NU UP EI PL ZR NA PE NC
0744:0003 8ED8          MOV     DS,AX
-t
AX=0744 BX=0000 CX=000F DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0744 IP=0005 NU UP EI PL ZR NA PE NC
0744:0005 B402          MOV     AH,02
-t
AX=0244 BX=0000 CX=000F DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0744 IP=0007 NU UP EI PL ZR NA PE NC
0744:0007 B241          MOV     DL,41
-t
AX=0244 BX=0000 CX=000F DX=0041 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0744 IP=0009 NU UP EI PL ZR NA PE NC
0744:0009 CD21          INT     21
-t
AAX=0241 BX=0000 CX=000F DX=0041 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0744 IP=000B NU UP EI PL ZR NA PE NC
0744:000B B44C          MOV     AH,4C
-t
AX=4C41 BX=0000 CX=000F DX=0041 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0744 IP=000D NU UP EI PL ZR NA PE NC
0744:000D CD21          INT     21
```

Problem 2 — Convert a Typed Letter to Uppercase

DATA SEGMENT

DATA ENDS

CODE SEGMENT

ASSUME CS:CODE, DS:DATA

START:

MOV AX,DATA

MOV DS,AX

MOV AH, 1 ;read a character

INT 21h

SUB AL, 20h ;lower to upper

MOV DL, AL

MOV AH,2 ;print a character

INT 21h

MOV AH,4CH

INT 21H

CODE ENDS

END START

```
-t
AX=0744 BX=0000 CX=0015 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0734 ES=0734 SS=0743 CS=0744 IP=0003 NU UP EI PL ZR NA PE NC
0744:0003 8ED8          MOV     DS,AX
-t
AX=0744 BX=0000 CX=0015 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0744 IP=0005 NU UP EI PL ZR NA PE NC
0744:0005 B401          MOV     AH,01
-t
AX=0144 BX=0000 CX=0015 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0744 IP=0007 NU UP EI PL ZR NA PE NC
0744:0007 CD21          INT     21
-t
AX=016D BX=0000 CX=0015 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0744 IP=0009 NU UP EI PL ZR NA PE NC
0744:0009 2C20          SUB     AL,20
-t
AX=014D BX=0000 CX=0015 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0744 IP=000B NU UP EI PL NZ NA PE NC
0744:000B 8AD0          MOV     DL,AL
-t
AX=014D BX=0000 CX=0015 DX=004D SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0744 IP=000D NU UP EI PL NZ NA PE NC
0744:000D B402          MOV     AH,02
```

```

-t
AX=024D BX=0000 CX=0015 DX=004D SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0744 IP=000F NU UP EI PL NZ NA PE NC
0744:000F CD21          INT     21
-t
MAX=024D BX=0000 CX=0015 DX=004D SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0744 IP=0011 NU UP EI PL NZ NA PE NC
0744:0011 B44C          MOV     AH,4C
-t
AX=4C4D BX=0000 CX=0015 DX=004D SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0744 IP=0013 NU UP EI PL NZ NA PE NC
0744:0013 CD21          INT     21
-t
Program terminated normally (004D)

```

Problem 3 — Display a Message

```

DATA SEGMENT
    MSG DB 'HELLO WORLD$'
DATA ENDS
CODE SEGMENT
ASSUME CS:CODE, DS:DATA
START:
    MOV AX,DATA
    MOV DS,AX
    MOV DX, OFFSET MSG
    MOV AH, 09H
    INT 21H
    MOV AH,4CH
    INT 21H
CODE ENDS
END START

```

```

-t
AX=0744 BX=0000 CX=0020 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0734 ES=0734 SS=0743 CS=0745 IP=0003 NU UP EI PL ZR NA PE NC
0745:0003 8ED8          MOV     DS,AX
-t
AX=0744 BX=0000 CX=0020 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0005 NU UP EI PL ZR NA PE NC
0745:0005 BA0000       MOV     DX,0000
-t
AX=0744 BX=0000 CX=0020 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0008 NU UP EI PL ZR NA PE NC
0745:0008 B409          MOV     AH,09

```

```

-t
AX=0944 BX=0000 CX=0020 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=000A NU UP EI PL ZR NA PE NC
0745:000A CD21          INT      21
-t
HELLO WORLDAX=0944 BX=0000 CX=0020 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=000C NU UP EI PL ZR NA PE NC
0745:000C B44C          MOV      AH,4C
-t
AX=4C44 BX=0000 CX=0020 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=000E NU UP EI PL ZR NA PE NC
0745:000E CD21          INT      21
-t

Program terminated normally (0044)

```

Problem 4 — Read and Display a String

DATA SEGMENT

```

MES1    DB 'Enter a string : $'
STR_BUF DB 255, 256 DUP(0)
NL       DB 0DH, 0AH, '$'
MES2    DB 'String entered is : $'

```

DATA ENDS

CODE SEGMENT

```
ASSUME CS:CODE, DS:DATA
```

START:

```

MOV AX, DATA
MOV DS, AX

```

```

MOV AH, 9
MOV DX, OFFSET MES1
INT 21H

```

```

MOV AH, 10
MOV DX, OFFSET STR_BUF
INT 21H

```

```
MOV SI, OFFSET STR_BUF+1
```

```

MOV CL, BYTE PTR [SI]
MOV CH, 0

```

```
INC SI
ADD SI, CX
```

```
MOV BYTE PTR [SI], '$'
```

```
MOV AH, 9
MOV DX, OFFSET NL
INT 21H
```

```
MOV AH, 9
MOV DX, OFFSET MES2
INT 21H
```

```
MOV AH, 9
MOV DX, OFFSET STR_BUF+2
INT 21H
```

```
MOV AX, 4C00H
INT 21H
```

```
CODE ENDS
END START
```

```
AX=0744 BX=0000 CX=016A DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0734 ES=0734 SS=0743 CS=0757 IP=0003 NU UP EI PL ZR NA PE NC
0757:0003 8ED8          MOV     DS,AX
-t
AX=0744 BX=0000 CX=016A DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0005 NU UP EI PL ZR NA PE NC
0757:0005 B409          MOV     AH,09
-t
AX=0944 BX=0000 CX=016A DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0007 NU UP EI PL ZR NA PE NC
0757:0007 BA0000        MOV     DX,0000
-t
AX=0944 BX=0000 CX=016A DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=000A NU UP EI PL ZR NA PE NC
0757:000A CD21          INT     21
-t
Enter a string : AX=0944 BX=0000 CX=016A DX=0000 SP=0000 BP=0000 SI=0000 DI
DS=0744 ES=0734 SS=0743 CS=0757 IP=000C NU UP EI PL ZR NA PE NC
0757:000C B40A          MOV     AH,0A
-t
AX=0A44 BX=0000 CX=016A DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=000E NU UP EI PL ZR NA PE NC
0757:000E BA1200        MOV     DX,0012
```

```

-t
AX=0A44 BX=0000 CX=016A DX=0012 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0011 NU UP EI PL ZR NA PE NC
0757:0011 CD21          INT      21
-t
AX=0A44 BX=0000 CX=016A DX=0012 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0013 NU UP EI PL ZR NA PE NC
0757:0013 BE1300        MOV      SI,0013
-t
AX=0A44 BX=0000 CX=016A DX=0012 SP=0000 BP=0000 SI=0013 DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0016 NU UP EI PL ZR NA PE NC
0757:0016 8A0C          MOV      CL,[SI]          DS:001
-t
AX=0A44 BX=0000 CX=0106 DX=0012 SP=0000 BP=0000 SI=0013 DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0018 NU UP EI PL ZR NA PE NC
0757:0018 B500          MOV      CH,00
-t
AX=0A44 BX=0000 CX=0006 DX=0012 SP=0000 BP=0000 SI=0013 DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=001A NU UP EI PL ZR NA PE NC
0757:001A 46           INC      SI
-t
AX=0A44 BX=0000 CX=0006 DX=0012 SP=0000 BP=0000 SI=0014 DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=001B NU UP EI PL NZ NA PE NC
0757:001B 03F1         ADD      SI,CX

```

```

AX=0A44 BX=0000 CX=0006 DX=0012 SP=0000 BP=0000 SI=001A DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=001D NU UP EI PL NZ NA PO NC
0757:001D C60424        MOV      BYTE PTR [SI],24          DS:001
-t
AX=0A44 BX=0000 CX=0006 DX=0012 SP=0000 BP=0000 SI=001A DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0020 NU UP EI PL NZ NA PO NC
0757:0020 B409          MOV      AH,09
-t
AX=0944 BX=0000 CX=0006 DX=0012 SP=0000 BP=0000 SI=001A DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0022 NU UP EI PL NZ NA PO NC
0757:0022 BA1301        MOV      DX,0113
-t
AX=0944 BX=0000 CX=0006 DX=0113 SP=0000 BP=0000 SI=001A DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0025 NU UP EI PL NZ NA PO NC
0757:0025 CD21          INT      21
-t
AX=0944 BX=0000 CX=0006 DX=0113 SP=0000 BP=0000 SI=001A DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0027 NU UP EI PL NZ NA PO NC
0757:0027 B409          MOV      AH,09
-t
AX=0944 BX=0000 CX=0006 DX=0113 SP=0000 BP=0000 SI=001A DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0029 NU UP EI PL NZ NA PO NC
0757:0029 BA1601        MOV      DX,0116

```

```

AX=0944 BX=0000 CX=0006 DX=0116 SP=0000 BP=0000 SI=001A DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=002C NU UP EI PL NZ NA PO NC
0757:002C CD21          INT     21
-t
String entered is : AX=0944 BX=0000 CX=0006 DX=0116 SP=0000 BP=0000 SI=001A
000
DS=0744 ES=0734 SS=0743 CS=0757 IP=002E NU UP EI PL NZ NA PO NC
0757:002E B409          MOV     AH,09
-t
AX=0944 BX=0000 CX=0006 DX=0116 SP=0000 BP=0000 SI=001A DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0030 NU UP EI PL NZ NA PO NC
0757:0030 BA1400        MOV     DX,0014
-t
AX=0944 BX=0000 CX=0006 DX=0014 SP=0000 BP=0000 SI=001A DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0033 NU UP EI PL NZ NA PO NC
0757:0033 CD21          INT     21
-t
mansviAX=0944 BX=0000 CX=0006 DX=0014 SP=0000 BP=0000 SI=001A DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0035 NU UP EI PL NZ NA PO NC
0757:0035 B8004C        MOV     AX,4C00
-t
AX=4C00 BX=0000 CX=0006 DX=0014 SP=0000 BP=0000 SI=001A DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0038 NU UP EI PL NZ NA PO NC
0757:0038 CD21          INT     21

```

```

C:\>cd tasm

C:\TASM>tasm demo.asm
Turbo Assembler  Version 3.0  Copyright (c) 1988, 1991 Borland International

Assembling file:   demo.asm
Error messages:    None
Warning messages:  None
Passes:            1
Remaining memory:  476k

C:\TASM>tlink demo.obj
Turbo Link  Version 3.0 Copyright (c) 1987, 1990 Borland International
Warning: No stack

C:\TASM>demo.exe
Enter a string : mansvi
String entered is : mansvi

```

Map file: where you data and code segment start and end it shows that.
Also give length of that range and cs: offset

C:\TASM>type demo.map

Start	Stop	Length	Name	Class
00000H	0012AH	0012BH	DATA	
00130H	00169H	0003AH	CODE	

Program entry point at 0013:0000
Warning: No stack

C:\DEBUG125>debug c:\tasm\demo.exe

-t 00130

AX=0744 BX=0000 CX=016A DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0734 ES=0734 SS=0743 CS=0757 IP=0003 NU UP EI PL ZR NA PE NC
0757:0003 8ED8 MOV DS,AX

AX=0744 BX=0000 CX=016A DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0005 NU UP EI PL ZR NA PE NC
0757:0005 B409 MOV AH,09

AX=0944 BX=0000 CX=016A DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0007 NU UP EI PL ZR NA PE NC
0757:0007 BA0000 MOV DX,0000

AX=0944 BX=0000 CX=016A DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=000A NU UP EI PL ZR NA PE NC
0757:000A CD21 INT 21

Enter a string : AX=0944 BX=0000 CX=016A DX=0000 SP=0000 BP=0000 SI=0000 DI=0000

DS=0744 ES=0734 SS=0743 CS=0757 IP=000C NU UP EI PL ZR NA PE NC
0757:000C B40A MOV AH,0A

AX=0A44 BX=0000 CX=016A DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=000E NU UP EI PL ZR NA PE NC
0757:000E BA1200 MOV DX,0012

AX=0A44 BX=0000 CX=016A DX=0012 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0011 NU UP EI PL ZR NA PE NC
0757:0011 CD21 INT 21

mansvi

```

AX=0944 BX=0000 CX=0006 DX=0113 SP=0000 BP=0000 SI=001A DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0029 NU UP EI PL NZ NA PO NC
0757:0029 BA1601          MOV     DX,0116
AX=0944 BX=0000 CX=0006 DX=0116 SP=0000 BP=0000 SI=001A DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=002C NU UP EI PL NZ NA PO NC
0757:002C CD21           INT     21
String entered is : AX=0944 BX=0000 CX=0006 DX=0116 SP=0000 BP=0000 SI=001A
000
DS=0744 ES=0734 SS=0743 CS=0757 IP=002E NU UP EI PL NZ NA PO NC
0757:002E B409          MOV     AH,09
AX=0944 BX=0000 CX=0006 DX=0116 SP=0000 BP=0000 SI=001A DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0030 NU UP EI PL NZ NA PO NC
0757:0030 BA1400          MOV     DX,0014
AX=0944 BX=0000 CX=0006 DX=0014 SP=0000 BP=0000 SI=001A DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0033 NU UP EI PL NZ NA PO NC
0757:0033 CD21           INT     21
mansviAX=0944 BX=0000 CX=0006 DX=0014 SP=0000 BP=0000 SI=001A DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0035 NU UP EI PL NZ NA PO NC
0757:0035 B8004C          MOV     AX,4C00
AX=4C00 BX=0000 CX=0006 DX=0014 SP=0000 BP=0000 SI=001A DI=0000
DS=0744 ES=0734 SS=0743 CS=0757 IP=0038 NU UP EI PL NZ NA PO NC
0757:0038 CD21           INT     21
Program terminated normally (0000)

```

Part B Programs

Problem 5 — Check Whether Two 16-bit Numbers Are Equal

DATA SEGMENT

```

NUM1 DW 1234H
NUM2 DW 1234H
MSG1 DB 'Equal$'
MSG2 DB 'Not Equal$'

```

DATA ENDS

CODE SEGMENT

```
ASSUME CS:CODE, DS:DATA
```

START:

```

MOV AX, DATA
MOV DS, AX

MOV AX, NUM1
CMP AX, NUM2
JNZ NOT_EQUAL

```

```

LEA DX, MSG1
MOV AH, 9
INT 21H
JMP DONE

```

```

NOT_EQUAL:
    LEA DX, MSG2
    MOV AH, 9
    INT 21H

```

```

DONE:
    MOV AH, 4CH
    INT 21H

```

```

CODE ENDS
END START

```

```

-t
AX=0744 BX=0000 CX=0043 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0734 ES=0734 SS=0743 CS=0746 IP=0003 NU UP EI PL ZR NA PE NC
0746:0003 8ED8          MOV     DS,AX
-t
AX=0744 BX=0000 CX=0043 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0746 IP=0005 NU UP EI PL ZR NA PE NC
0746:0005 A10000       MOV     AX,[0000]          DS:0000=
-t
AX=1234 BX=0000 CX=0043 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0746 IP=0008 NU UP EI PL ZR NA PE NC
0746:0008 3B060200     CMP     AX,[0002]          DS:0002=
-t
AX=1234 BX=0000 CX=0043 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0746 IP=000C NU UP EI PL ZR NA PE NC
0746:000C 750A        JNZ     0018
-t
AX=1234 BX=0000 CX=0043 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0746 IP=000E NU UP EI PL ZR NA PE NC
0746:000E BA0400       MOV     DX,0004
-t
AX=1234 BX=0000 CX=0043 DX=0004 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0746 IP=0011 NU UP EI PL ZR NA PE NC
0746:0011 B409        MOV     AH,09

```

```

-t
AX=0934 BX=0000 CX=0043 DX=0004 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0746 IP=0013 NU UP EI PL ZR NA PE NC
0746:0013 CD21          INT     21
-t
Equal AX=0934 BX=0000 CX=0043 DX=0004 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0746 IP=0015 NU UP EI PL ZR NA PE NC
0746:0015 EB08          JMP     001F
-t
AX=0934 BX=0000 CX=0043 DX=0004 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0746 IP=001F NU UP EI PL ZR NA PE NC
0746:001F B44C          MOV     AH,4C
-t
AX=4C34 BX=0000 CX=0043 DX=0004 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0746 IP=0021 NU UP EI PL ZR NA PE NC
0746:0021 CD21          INT     21
-t
Program terminated normally (0034)

```

Problem 6 — Minimum of Two 16-bit Numbers

DATA SEGMENT

NUM1 DW 1234H

NUM2 DW 2345H

MINIMUM DW ?

DATA ENDS

CODE SEGMENT

ASSUME CS:CODE, DS:DATA

START:

MOV AX, DATA

MOV DS, AX

MOV AX, NUM1

CMP AX, NUM2

JC NUM1_SMALL

MOV AX, NUM2

NUM1_SMALL:

MOV MINIMUM, AX

DONE:

```
MOV AH, 4CH
INT 21H
```

```
CODE ENDS
END START
```

```
-t
AX=0744 BX=0000 CX=0028 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0734 ES=0734 SS=0743 CS=0745 IP=0003 NU UP EI PL ZR NA PE NC
0745:0003 8ED8          MOV     DS,AX
-t
AX=0744 BX=0000 CX=0028 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0005 NU UP EI PL ZR NA PE NC
0745:0005 A10000       MOV     AX,[0000]          DS:0000=
-t
AX=1234 BX=0000 CX=0028 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0008 NU UP EI PL ZR NA PE NC
0745:0008 3B060200     CMP     AX,[0002]          DS:0002=
-t
AX=1234 BX=0000 CX=0028 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=000C NU UP EI NG NZ AC PO CY
0745:000C 7203        JB      0011
-t
AX=1234 BX=0000 CX=0028 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0011 NU UP EI NG NZ AC PO CY
0745:0011 A30400       MOV     [0004],AX          DS:0004=
-t
AX=1234 BX=0000 CX=0028 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0014 NU UP EI NG NZ AC PO CY
0745:0014 B44C        MOV     AH,4C
```

Problem 7 — Odd or Even

```
DATA SEGMENT
    NUM DW 1235H
    ODD_MSG DB 'Odd$'
    EVEN_MSG DB 'Even$'
DATA ENDS
CODE SEGMENT
ASSUME CS:CODE, DS:DATA
START:
    MOV AX, DATA
    MOV DS, AX

    MOV AX, NUM
    AND AX, 0001H
    CMP AX, 1
```

JE ODD

LEA DX, EVEN_MSG
MOV AH, 9
INT 21H
JMP DONE

ODD:

LEA DX, ODD_MSG
MOV AH, 9
INT 21H

DONE:

MOV AH, 4CH
INT 21H

CODE ENDS

END START

```
-t
AX=0744 BX=0000 CX=0035 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0734 ES=0734 SS=0743 CS=0745 IP=0003 NU UP EI PL ZR NA PE NC
0745:0003 8ED8          MOV     DS,AX
-t
AX=0744 BX=0000 CX=0035 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0005 NU UP EI PL ZR NA PE NC
0745:0005 A10000       MOV     AX,[0000]          DS:0000=
-t
AX=1235 BX=0000 CX=0035 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0008 NU UP EI PL ZR NA PE NC
0745:0008 250100       AND     AX,0001
-t
AX=0001 BX=0000 CX=0035 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=000B NU UP EI PL NZ NA PO NC
0745:000B 3D0100       CMP     AX,0001
-t
AX=0001 BX=0000 CX=0035 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=000E NU UP EI PL ZR NA PE NC
0745:000E 740A        JZ      001A
-t
AX=0001 BX=0000 CX=0035 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=001A NU UP EI PL ZR NA PE NC
0745:001A BA0200       MOV     DX,0002
```

```

-t
AX=0001 BX=0000 CX=0035 DX=0002 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=001D NU UP EI PL ZR NA PE NC
0745:001D B409          MOV     AH,09
-t
AX=0901 BX=0000 CX=0035 DX=0002 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=001F NU UP EI PL ZR NA PE NC
0745:001F CD21          INT     21
-t
0ddAX=0901 BX=0000 CX=0035 DX=0002 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0021 NU UP EI PL ZR NA PE NC
0745:0021 B44C          MOV     AH,4C
-t
AX=4C01 BX=0000 CX=0035 DX=0002 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0023 NU UP EI PL ZR NA PE NC
0745:0023 CD21          INT     21
-t
Program terminated normally (0001)

```

Problem 8 — Minimum from a Block of Numbers (With a Loop)

DATA SEGMENT

ARR DW 45H, 12H, 78H, 09H, 34H, 67H

MINIMUM DW ?

DATA ENDS

CODE SEGMENT

ASSUME CS:CODE, DS:DATA

START:

MOV AX, DATA

MOV DS, AX

MOV SI, OFFSET ARR

MOV AX, [SI]

ADD SI, 2

MOV CX, 5

NEXT:

CMP AX, [SI]

JB NO_CHANGE

MOV AX, [SI]

NO_CHANGE:

ADD SI, 2

LOOP NEXT

MOV MINIMUM, AX

DONE:

MOV AH, 4CH

INT 21H

CODE ENDS

END START

```
-t
AX=0744 BX=0000 CX=0032 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0734 ES=0734 SS=0743 CS=0745 IP=0003 NU UP EI PL ZR NA PE NC
0745:0003 8ED8          MOV     DS,AX
-t
AX=0744 BX=0000 CX=0032 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0005 NU UP EI PL ZR NA PE NC
0745:0005 BE0000      MOV     SI,0000
-t
AX=0744 BX=0000 CX=0032 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0008 NU UP EI PL ZR NA PE NC
0745:0008 8B04          MOV     AX,[SI]          DS:0000=
-t
AX=0045 BX=0000 CX=0032 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=000A NU UP EI PL ZR NA PE NC
0745:000A 83C602      ADD     SI,+02
-t
AX=0045 BX=0000 CX=0032 DX=0000 SP=0000 BP=0000 SI=0002 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=000D NU UP EI PL NZ NA PO NC
0745:000D B90500      MOV     CX,0005
-t
AX=0045 BX=0000 CX=0005 DX=0000 SP=0000 BP=0000 SI=0002 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0010 NU UP EI PL NZ NA PO NC
0745:0010 3B04          CMP     AX,[SI]          DS:0002=
-
```

```

-t
AX=0045 BX=0000 CX=0005 DX=0000 SP=0000 BP=0000 SI=0002 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0012 NU UP EI PL NZ NA PE NC
0745:0012 7202                JB      0016
-t
AX=0045 BX=0000 CX=0005 DX=0000 SP=0000 BP=0000 SI=0002 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0014 NU UP EI PL NZ NA PE NC
0745:0014 8B04                MOV     AX,[SI]          DS:0002=
-t
AX=0012 BX=0000 CX=0005 DX=0000 SP=0000 BP=0000 SI=0002 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0016 NU UP EI PL NZ NA PE NC
0745:0016 83C602              ADD     SI,+02
-t
AX=0012 BX=0000 CX=0005 DX=0000 SP=0000 BP=0000 SI=0004 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0019 NU UP EI PL NZ NA PO NC
0745:0019 E2F5                LOOPW  0010
-t
AX=0012 BX=0000 CX=0004 DX=0000 SP=0000 BP=0000 SI=0004 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0010 NU UP EI PL NZ NA PO NC
0745:0010 3B04                CMP     AX,[SI]          DS:0004=
-t
AX=0012 BX=0000 CX=0004 DX=0000 SP=0000 BP=0000 SI=0004 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0012 NU UP EI NG NZ AC PE CY
0745:0012 7202                JB      0016

```

```

-t
AX=0012 BX=0000 CX=0004 DX=0000 SP=0000 BP=0000 SI=0004 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0016 NU UP EI NG NZ AC PE CY
0745:0016 83C602              ADD     SI,+02
-t
AX=0012 BX=0000 CX=0004 DX=0000 SP=0000 BP=0000 SI=0006 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0019 NU UP EI PL NZ NA PE NC
0745:0019 E2F5                LOOPW  0010
-t
AX=0012 BX=0000 CX=0003 DX=0000 SP=0000 BP=0000 SI=0006 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0010 NU UP EI PL NZ NA PE NC
0745:0010 3B04                CMP     AX,[SI]          DS:0006=
-t
AX=0012 BX=0000 CX=0003 DX=0000 SP=0000 BP=0000 SI=0006 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0012 NU UP EI PL NZ AC PE NC
0745:0012 7202                JB      0016
-t
AX=0012 BX=0000 CX=0003 DX=0000 SP=0000 BP=0000 SI=0006 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0014 NU UP EI PL NZ AC PE NC
0745:0014 8B04                MOV     AX,[SI]          DS:0006=
-t
AX=0009 BX=0000 CX=0003 DX=0000 SP=0000 BP=0000 SI=0006 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0016 NU UP EI PL NZ AC PE NC
0745:0016 83C602              ADD     SI,+02

```

```

-t
AX=0009 BX=0000 CX=0003 DX=0000 SP=0000 BP=0000 SI=0008 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0019 NU UP EI PL NZ NA PO NC
0745:0019 E2F5          LOOPW  0010
-t
AX=0009 BX=0000 CX=0002 DX=0000 SP=0000 BP=0000 SI=0008 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0010 NU UP EI PL NZ NA PO NC
0745:0010 3B04          CMP     AX,[SI]          DS:0008=
-t
AX=0009 BX=0000 CX=0002 DX=0000 SP=0000 BP=0000 SI=0008 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0012 NU UP EI NG NZ NA PO CY
0745:0012 7202          JB      0016
-t
AX=0009 BX=0000 CX=0002 DX=0000 SP=0000 BP=0000 SI=0008 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0016 NU UP EI NG NZ NA PO CY
0745:0016 83C602        ADD     SI,+02
-t
AX=0009 BX=0000 CX=0002 DX=0000 SP=0000 BP=0000 SI=000A DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0019 NU UP EI PL NZ NA PE NC
0745:0019 E2F5          LOOPW  0010
-t
AX=0009 BX=0000 CX=0001 DX=0000 SP=0000 BP=0000 SI=000A DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0010 NU UP EI PL NZ NA PE NC
0745:0010 3B04          CMP     AX,[SI]          DS:000A=
-

```

```

-t
AX=0009 BX=0000 CX=0001 DX=0000 SP=0000 BP=0000 SI=000A DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0012 NU UP EI NG NZ NA PO CY
0745:0012 7202          JB      0016
-t
AX=0009 BX=0000 CX=0001 DX=0000 SP=0000 BP=0000 SI=000A DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0016 NU UP EI NG NZ NA PO CY
0745:0016 83C602        ADD     SI,+02
-t
AX=0009 BX=0000 CX=0001 DX=0000 SP=0000 BP=0000 SI=000C DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0019 NU UP EI PL NZ NA PE NC
0745:0019 E2F5          LOOPW  0010
-t
AX=0009 BX=0000 CX=0000 DX=0000 SP=0000 BP=0000 SI=000C DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=001B NU UP EI PL NZ NA PE NC
0745:001B A30C00        MOV     [000C],AX          DS:000C=
-t
AX=0009 BX=0000 CX=0000 DX=0000 SP=0000 BP=0000 SI=000C DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=001E NU UP EI PL NZ NA PE NC
0745:001E B44C          MOV     AH,4C
-t
AX=4C09 BX=0000 CX=0000 DX=0000 SP=0000 BP=0000 SI=000C DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0020 NU UP EI PL NZ NA PE NC
0745:0020 CD21          INT     21
-t
-

```

Problem 9 — Factorial of an 8-bit Number

```
DATA SEGMENT
    NUM DB 5
    FACT DW ?
DATA ENDS
CODE SEGMENT
ASSUME CS:CODE, DS:DATA
START:
    MOV AX, DATA
    MOV DS, AX

    MOV CL, NUM
    MOV CH, 0

    MOV AX, 1
    MOV BX, 1

NEXT:
    MUL BX
    INC BX
    LOOP NEXT

    MOV FACT, AX

DONE:
    MOV AH, 4CH
    INT 21H

CODE ENDS
END START
```

```

-t
AX=0744 BX=0000 CX=002D DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0734 ES=0734 SS=0743 CS=0745 IP=0003 NU UP EI PL ZR NA PE NC
0745:0003 8ED8          MOV     DS,AX
-t
AX=0744 BX=0000 CX=002D DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0005 NU UP EI PL ZR NA PE NC
0745:0005 8A0E0000      MOV     CL,[0000]
DS:0000=05
-t
AX=0744 BX=0000 CX=0005 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0009 NU UP EI PL ZR NA PE NC
0745:0009 B500          MOV     CH,00
-t
AX=0744 BX=0000 CX=0005 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=000B NU UP EI PL ZR NA PE NC
0745:000B B80100      MOV     AX,0001
-t
AX=0001 BX=0000 CX=0005 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=000E NU UP EI PL ZR NA PE NC
0745:000E BB0100      MOV     BX,0001
-t
AX=0001 BX=0001 CX=0005 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0011 NU UP EI PL ZR NA PE NC
0745:0011 F7E3          MUL     BX
-

```

```

-t
AX=0002 BX=0003 CX=0004 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0014 NU UP EI PL NZ NA PE NC
0745:0014 E2FB          LOOPW  0011
-t
AX=0002 BX=0003 CX=0003 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0011 NU UP EI PL NZ NA PE NC
0745:0011 F7E3          MUL     BX
-t
AX=0006 BX=0003 CX=0003 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0013 NU UP EI PL NZ NA PE NC
0745:0013 43           INC     BX
-t
AX=0006 BX=0004 CX=0003 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0014 NU UP EI PL NZ NA PO NC
0745:0014 E2FB          LOOPW  0011
-t
AX=0006 BX=0004 CX=0002 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0011 NU UP EI PL NZ NA PO NC
0745:0011 F7E3          MUL     BX
-t
AX=0018 BX=0004 CX=0002 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0013 NU UP EI PL NZ NA PO NC
0745:0013 43           INC     BX
-

```

```

-t
AX=0001 BX=0001 CX=0005 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0011 NU UP EI PL ZR NA PE NC
0745:0011 F7E3          MUL     BX
-t
AX=0001 BX=0001 CX=0005 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0013 NU UP EI PL NZ NA PE NC
0745:0013 43           INC     BX
-t
AX=0001 BX=0002 CX=0005 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0014 NU UP EI PL NZ NA PO NC
0745:0014 E2FB          LOOPW  0011
-t
AX=0001 BX=0002 CX=0004 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0011 NU UP EI PL NZ NA PO NC
0745:0011 F7E3          MUL     BX
-t
AX=0002 BX=0002 CX=0004 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0013 NU UP EI PL NZ NA PO NC
0745:0013 43           INC     BX
-t
AX=0002 BX=0003 CX=0004 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0014 NU UP EI PL NZ NA PE NC
0745:0014 E2FB          LOOPW  0011
-t

```

```

-t
AX=0018 BX=0004 CX=0002 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0013 NU UP EI PL NZ NA PO NC
0745:0013 43          INC      BX
-t
AX=0018 BX=0005 CX=0002 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0014 NU UP EI PL NZ NA PE NC
0745:0014 E2FB      LOOPW    0011
-t
AX=0018 BX=0005 CX=0001 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0011 NU UP EI PL NZ NA PE NC
0745:0011 F7E3      MUL      BX
-t
AX=0078 BX=0005 CX=0001 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0013 NU UP EI PL NZ NA PE NC
0745:0013 43          INC      BX
-t
AX=0078 BX=0006 CX=0001 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0014 NU UP EI PL NZ NA PE NC
0745:0014 E2FB      LOOPW    0011
-t
AX=0078 BX=0006 CX=0000 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0016 NU UP EI PL NZ NA PE NC
0745:0016 A30100    MOV      [0001],AX          DS:0001=0000

```

```

0745:0011 F7E3      MUL      BX
-t
AX=0078 BX=0005 CX=0001 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0013 NU UP EI PL NZ NA PE NC
0745:0013 43          INC      BX
-t
AX=0078 BX=0006 CX=0001 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0014 NU UP EI PL NZ NA PE NC
0745:0014 E2FB      LOOPW    0011
-t
AX=0078 BX=0006 CX=0000 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0016 NU UP EI PL NZ NA PE NC
0745:0016 A30100    MOV      [0001],AX          DS:0001=0000
-t
AX=0078 BX=0006 CX=0000 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0019 NU UP EI PL NZ NA PE NC
0745:0019 B44C      MOV      AH,4C
-t
AX=4C78 BX=0006 CX=0000 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=001B NU UP EI PL NZ NA PE NC
0745:001B CD21      INT      21

```

Program terminated normally (0078)

Problem 10 — Check Whether All Numbers in an Array Are the Same

DATA SEGMENT

ARR DW 25H, 25H, 25H, 25H, 25H

MSG1 DB 'Same\$'

MSG2 DB 'Not same\$'

DATA ENDS

CODE SEGMENT

ASSUME CS:CODE, DS:DATA

START:

MOV AX, DATA

MOV DS, AX

MOV SI, OFFSET ARR

MOV AX, [SI]

ADD SI, 2

MOV CX, 4

NEXT:

CMP AX, [SI]

ADD SI, 2

LOOPE NEXT

CMP CX, 0

JE SAME

LEA DX, MSG2

MOV AH, 9

INT 21H

JMP DONE

SAME:

LEA DX, MSG1

MOV AH, 9

INT 21H

DONE:

MOV AH, 4CH

INT 21H

CODE ENDS

END START

```
-t
AX=0744 BX=0000 CX=002D DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0734 ES=0734 SS=0743 CS=0745 IP=0003 NU UP EI PL ZR NA PE NC
0745:0003 8ED8          MOV     DS,AX
-t
AX=0744 BX=0000 CX=002D DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0005 NU UP EI PL ZR NA PE NC
0745:0005 8A0E0000      MOV     CL,[0000]
DS:0000=05
-t
AX=0744 BX=0000 CX=0005 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0009 NU UP EI PL ZR NA PE NC
0745:0009 B500          MOV     CH,00
-t
AX=0744 BX=0000 CX=0005 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=000B NU UP EI PL ZR NA PE NC
0745:000B B80100      MOV     AX,0001
-t
AX=0001 BX=0000 CX=0005 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=000E NU UP EI PL ZR NA PE NC
0745:000E BB0100      MOV     BX,0001
-t
AX=0001 BX=0001 CX=0005 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0011 NU UP EI PL ZR NA PE NC
0745:0011 F7E3          MUL     BX
-
-t
AX=0001 BX=0001 CX=0005 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0011 NU UP EI PL ZR NA PE NC
0745:0011 F7E3          MUL     BX
-t
AX=0001 BX=0001 CX=0005 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0013 NU UP EI PL NZ NA PE NC
0745:0013 43            INC     BX
-t
AX=0001 BX=0002 CX=0005 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0014 NU UP EI PL NZ NA PO NC
0745:0014 E2FB          LOOPW  0011
-t
AX=0001 BX=0002 CX=0004 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0011 NU UP EI PL NZ NA PO NC
0745:0011 F7E3          MUL     BX
-t
AX=0002 BX=0002 CX=0004 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0013 NU UP EI PL NZ NA PO NC
0745:0013 43            INC     BX
-t
AX=0002 BX=0003 CX=0004 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0014 NU UP EI PL NZ NA PE NC
0745:0014 E2FB          LOOPW  0011
-
```

```

-t
AX=0002 BX=0003 CX=0004 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0014 NU UP EI PL NZ NA PE NC
0745:0014 E2FB          LOOPW  0011
-t
AX=0002 BX=0003 CX=0003 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0011 NU UP EI PL NZ NA PE NC
0745:0011 F7E3          MUL     BX
-t
AX=0006 BX=0003 CX=0003 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0013 NU UP EI PL NZ NA PE NC
0745:0013 43           INC     BX
-t
AX=0006 BX=0004 CX=0003 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0014 NU UP EI PL NZ NA PO NC
0745:0014 E2FB          LOOPW  0011
-t
AX=0006 BX=0004 CX=0002 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0011 NU UP EI PL NZ NA PO NC
0745:0011 F7E3          MUL     BX
-t
AX=0018 BX=0004 CX=0002 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0013 NU UP EI PL NZ NA PO NC
0745:0013 43           INC     BX

```

```

-t
AX=0018 BX=0004 CX=0002 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0013 NU UP EI PL NZ NA PO NC
0745:0013 43           INC     BX
-t
AX=0018 BX=0005 CX=0002 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0014 NU UP EI PL NZ NA PE NC
0745:0014 E2FB          LOOPW  0011
-t
AX=0018 BX=0005 CX=0001 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0011 NU UP EI PL NZ NA PE NC
0745:0011 F7E3          MUL     BX
-t
AX=0078 BX=0005 CX=0001 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0013 NU UP EI PL NZ NA PE NC
0745:0013 43           INC     BX
-t
AX=0078 BX=0006 CX=0001 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0014 NU UP EI PL NZ NA PE NC
0745:0014 E2FB          LOOPW  0011
-t
AX=0078 BX=0006 CX=0000 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0016 NU UP EI PL NZ NA PE NC
0745:0016 A30100        MOV     [0001],AX

```

DS:0001=0000

```

0745:0011 F7E3          MUL     BX
-t
AX=0078 BX=0005 CX=0001 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0013 NU UP EI PL NZ NA PE NC
0745:0013 43          INC     BX
-t
AX=0078 BX=0006 CX=0001 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0014 NU UP EI PL NZ NA PE NC
0745:0014 E2FB      LOOPW    0011
-t
AX=0078 BX=0006 CX=0000 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0016 NU UP EI PL NZ NA PE NC
0745:0016 A30100     MOV     [0001],AX          DS:0001=0000
-t
AX=0078 BX=0006 CX=0000 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=0019 NU UP EI PL NZ NA PE NC
0745:0019 B44C     MOV     AH,4C
-t
AX=4C78 BX=0006 CX=0000 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0744 ES=0734 SS=0743 CS=0745 IP=001B NU UP EI PL NZ NA PE NC
0745:001B CD21     INT     21
-t

```